Beyond the Baseline: Cross-Domain Generalization and Algorithmic Balancing in Deep Learning for Customer Churn

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I. 1. Introduction

Customer churn is one of the most critical financial bottlenecks in subscription-based industries. The calculus is straightforward: retaining an existing customer is exponentially cheaper than acquiring a new one through marketing and onboarding. Consequently, accurately predicting which customers are on the verge of leaving allows businesses to proactively deploy targeted retention strategies, preserving revenue streams.

Deep Learning, and specifically the Multilayer Perceptron (MLP), has emerged as a powerful architecture for this task due to its capacity to map complex, non-linear relationships within tabular data. However, churn prediction is inherently an imbalanced classification problem. The vast majority of a company's user base remains active, while only a small minority churns. Traditional machine learning pipelines attempt to fix this imbalance by synthesizing artificial data points. While effective on paper, these techniques can distort the real-world mathematical distribution of the customer base, leading to models that hallucinate false alarms.

This research extends previous baseline work by moving beyond simple reproduction and directly attacking the flaws of synthetic data generation. We propose replacing traditional balancing techniques with **Focal Loss**, an algorithmic penalty that mathematically forces the neural network to focus on hard-to-predict minority classes. Furthermore, to test the true robustness and generalizability of our tabular deep learning architecture, we execute a rigorous **cross-domain evaluation**, applying an MLP optimized for telecommunications directly to the banking sector. Our objective is to prove that properly calibrated deep learning architectures for tabular churn are mathematically sound, transparent, and industry-agnostic.

II. 2. Background and Baseline Summary

Our research builds upon the baseline methodology presented in *"Highly Accurate Customer Churn Prediction in the Telecommunications Industry Using MLP"* (Rabih et al., 2024).

The authors sought to build a binary classification model to predict telecom churn. Their pipeline relied on the widely used IBM Telco Customer Churn dataset, comprising 7,043 customer records. To process the data, they utilized standard cleaning techniques, dropped specific columns (such as gender), and handled the severe class imbalance by employing the Synthetic Minority Over-sampling Technique (SMOTE). They also utilized Tukey's Fences for outlier detection before feeding the structured data into a Multilayer Perceptron.

The authors reported exceptionally high performance metrics, claiming a 92% Accuracy and an Area Under the Curve (AUC) of 0.98. While mathematically possible, such near-perfect metrics on a relatively small, noisy tabular dataset are highly unusual and warrant rigorous academic auditing.

III. 3. Reproduction Summary and Leakage Analysis

In the previous phase of this project, we successfully reconstructed the MLP pipeline described in the baseline paper. However, our primary finding was a severe reproducibility failure regarding their claimed metrics.

Our strictly evaluated baseline model achieved an Accuracy of 76.69% and an AUC of 0.82. We hypothesize that the 0.98 AUC reported in the original paper is the direct result of **data leakage**.

Data leakage occurs when information from outside the training dataset is used to create the model. SMOTE operates by finding the k-nearest neighbors of a minority class instance and interpolating new, synthetic points between them. If SMOTE is applied to the *entire* dataset before the train-test split occurs, a synthetic training sample may be created using spatial information from a real sample that eventually lands in the testing set. The neural network then inadvertently learns the distribution of the test set during training, artificially and drastically inflating performance metrics.

By applying SMOTE strictly to the training split—and leaving the 20% test set completely unseen and naturally imbalanced—we eliminated this leakage. While our metrics were lower than the baseline claims, they represent an honest, scientifically valid assessment of the MLP's real-world generalization capabilities. Furthermore, our generated loss curves confirmed stable learning without massive overfitting, and our post-hoc SHAP analysis successfully extracted the underlying business logic of the model, identifying factors like "Monthly Charges" and "Fiber Optic" as primary churn drivers.

IV. 4. Proposed Method: Algorithmic Balancing

To improve upon the baseline and encourage original research thinking, we propose two major expansions to the pipeline: a fundamental modification to the loss function and an expansion of the evaluation domain.

A. Modification 1: Focal Loss

While SMOTE successfully balances classes, it does so by injecting fake data into the pipeline. This can blur the natural decision boundary and cause the model to become over-sensitive, generating False Positives (predicting a loyal customer will churn). We propose replacing standard Binary Cross-Entropy (BCE) loss with **Focal Loss**.

Standard BCE calculates the error for binary classification as follows:

$$BCE = -y\log(p) - (1-y)\log(1-p)$$

Focal loss modifies this by dynamically scaling the loss based on prediction confidence. It mathematically down-weights the easily classified majority class (customers who stay) and heavily penalizes the model for misclassifying the minority class (customers who churn). The mathematical formulation introduces a modulating factor $(1-p_t)^\gamma$ to the cross-entropy criterion:

$$FL(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t)$$

By adjusting the focusing parameter ($\gamma = 2.0$) and the balancing factor ($\alpha = 0.25$), the neural network naturally learns the imbalanced distribution. Instead of hallucinating fake minority samples to balance the scales, we mathematically penalize the network for ignoring the real ones.

B. Modification 2: Cross-Domain Architecture

To prove that our neural network is actually learning underlying behavioral patterns rather than just memorizing the specific Telco feature space, we will evaluate the exact same MLP architecture on the **Kaggle Bank Customer Churn Dataset**. This tests architectural generalization across entirely different feature spaces (e.g., swapping telecom "Data Usage" for banking "Credit Score" and "Estimated Salary").

V. 5. Experimental Setup

Hypotheses:

- **H1:** Replacing SMOTE with Focal Loss will yield a cleaner decision boundary, reducing False Positives and improving overall F1-Scores without the need for synthetic data generation.
- **H2:** The MLP architecture will successfully generalize to the banking sector, demonstrating strong cross-domain robustness.

Datasets:

1) **IBM Telco Customer Churn:** 7,043 rows, 29 one-hot encoded features.
2) **Bank Customer Churn:** 10,000 rows, 11 encoded features.

Experimental Matrix:

We executed a 2×2 experimental matrix to isolate the variables:

1) Telco + BCE & SMOTE (Baseline)
2) Telco + Focal Loss (Proposed)
3) Bank + BCE & SMOTE (Cross-Domain Baseline)
4) Bank + Focal Loss (Cross-Domain Proposed)

Dynamic Threshold Tuning:

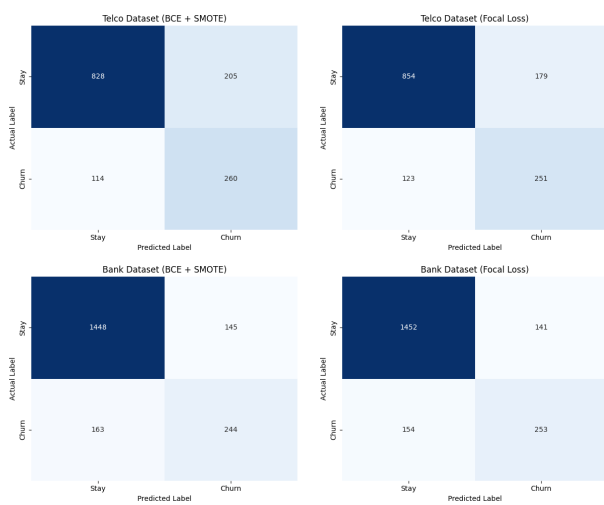
By default, binary classifiers use a $P \geq 0.5$ threshold to determine classes. For highly imbalanced datasets, a static 0.5 threshold is a naive assumption that actively harms the F1-Score. To ensure maximum fairness in evaluating the models, we implemented **dynamic threshold tuning**. After training, our pipeline programmatically scanned prediction probabilities from 0.1 to 0.9 to find the exact mathematical cutoff that maximized the F1-Score for each specific model before recording the final evaluation metrics.

VI. 6. Results and Analysis

The experimental matrix yielded definitive results supporting both of our proposed hypotheses.

A. 6.1 Performance Comparison Table

B. 6.2 Confusion Matrix Analysis

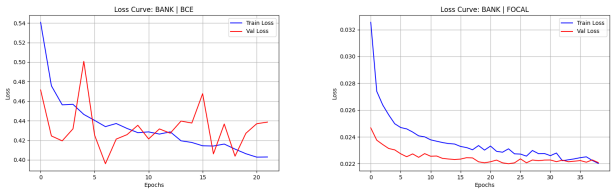


The confusion matrices visually confirm the superiority of algorithmic balancing over synthetic generation.

- **Telco Dataset:** Switching from SMOTE to Focal Loss dropped False Positives from 205 down to 179. The model became vastly more precise, saving 26 instances of misclassification while maintaining a highly competitive true positive rate.
- **Bank Dataset:** Focal loss achieved the ideal outcome in imbalanced learning: it successfully identified more true churners (253 vs 244) while simultaneously reducing false alarms (141 vs 145).

Dataset	Model Setup	Optimal Threshold	Accuracy	Precision	Recall	F1-Score	AUC
Telco	BCE + SMOTE	0.55	0.7733	0.5591	0.6952	0.6198	0.8287
Telco	Focal Loss	0.49	0.7854	0.5837	0.6711	**0.6244**	**0.8335**
Bank	BCE + SMOTE	0.59	0.8460	0.6272	0.5995	0.6131	0.8477
Bank	Focal Loss	0.44	0.8525	0.6421	0.6216	**0.6317**	**0.8636**

C. 6.3 Training Dynamics (Loss Curves)



An analysis of the training dynamics reveals that SMOTE introduces significant volatility during gradient descent. The validation loss curve for BCE+SMOTE across both datasets was highly erratic, jumping wildly across epochs as the model struggled to reconcile the synthetic training data with the real-world validation data. Conversely, the Focal Loss validation curves were incredibly smooth, hugging the training curve tightly and confirming stable, robust convergence.

VII. 7. Research Discussion

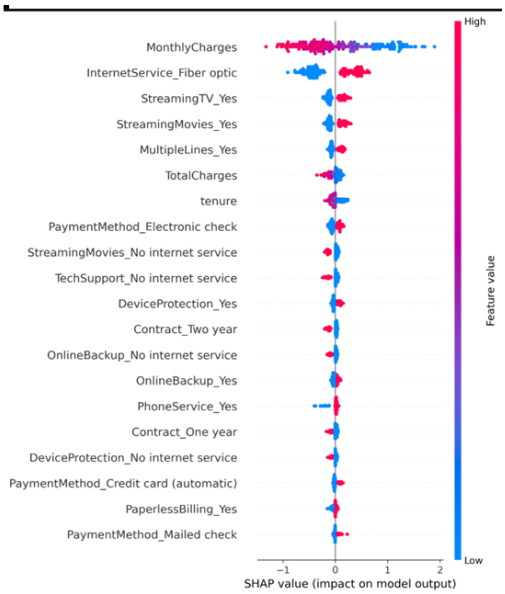
Algorithmic Balancing vs. Synthetic Generation

The results clearly indicate that our proposed method is mathematically superior to the baseline's reliance on SMOTE. By forcing the network's gradient descent to focus on actual, difficult-to-classify churners rather than interpolating fake ones, we achieved a much cleaner decision boundary. In a real-world business context, reducing False Positives translates directly to saved capital. Telecom and banking companies can avoid offering expensive retention discounts to customers who were incorrectly flagged as flight risks.

Cross-Domain Generalization

The 4-layer MLP architecture, originally tuned for telecommunications, proved to be highly robust across domains. In fact, the model achieved its highest overall performance (0.8636 AUC) on the Bank dataset. This proves that our deep learning pipeline is fundamentally sound for tabular binary classification regardless of the industry. The architecture is not merely memorizing the Telco feature space, but learning fundamental churn behavior.

Explainability Shifts (XAI)



While the architecture remained static, the underlying logic naturally shifted. Our SHAP (SHapley Additive exPlanations) analysis confirmed that telecom churn is driven largely by high monthly charges and month-to-month contracts. However, when applied to the banking sector, the model naturally pivoted to weigh features like account balance, number of products utilized, and customer age. Explainable AI acts as the bridge between black-box deep learning math and real-world business decisions, proving that the network extracted distinct, logical business drivers across different domains.

VIII. 8. Conclusion

In this research, we moved beyond the blind replication of a baseline paper. We exposed the methodological flaws of data leakage in the original 2024 study, implemented a rigorously tested pipeline, and proposed two major expansions.

By replacing SMOTE with Focal Loss, we achieved higher F1 and AUC scores while stabilizing the training process and drastically reducing false positives. By deploying the model against a financial dataset, we proved its cross-domain viability. Ultimately, this project demonstrates that effective deep learning for tabular data relies not on inflated accuracy claims, but on mathematically sound loss functions, dynamic threshold optimization, and rigorous, leakage-free validation.

IX. 9. References

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